

UNIVERSITY OF DAR ES SALAAM



COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGIES

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

IS335 / CS498: Final Year Project Report – End of Semester One

Project Title:	Property Management System		
	Neema Moses Mpangile	2022-04-08286	CEIT
Student Name:	Nuh Saidi Buhero	2022-04-00886	CEIT
	Hubert Ayubu Bahati	2022-04-00613	CEIT
	Hud Saidi Buhero	2022-04-00887	CEIT
Supervisor Name:	Dr. Daudi Charles		
Supervisor Signature			

TABLE OF CONTENTS

COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGIES.....	1
LIST OF ABBREVIATION.....	4
CHAPTER ONE.....	5
INTRODUCTION	5
1.1. General Introduction.....	5
1.2. Statement of the Problem	5
1.3. Objective	6
1.3.1. Main Objectives	6
1.3.2. Specific Objectives.....	6
1.4. Significance of the Project	7
1.5. Project Scope.....	7
1.6. Organization of the report	8
CHAPTER TWO	9
LITERATURE REVIEW	9
2.1. Overview of Property Management Systems	9
2.2. Tenant Onboarding and Identity Verification: Manual Processes vs Digital Integration.....	10
2.3. Financial Management and Payments: Foreign Gateways vs Tanzanian Payment Ecosystem	10
2.4. Maintenance, Communication, and Operational Systems.....	11
2.5. Related Works.....	11
Figure 1: House Rental Process via Middleman at Current residential processes.	12
2.6. Weaknesses of Existing Property Management Systems	12
CHAPTER THREE	14
METHODOLOGY	14
3.1. Research Design and Approach.....	14
3.2. System Development Methodology	14
Figure 2: Agile Development workflow	14
3.2.2 Scrum Agile Framework	15
CHAPTER 4.....	16
PROPERTY MANAGEMENT SYSTEM: SYSTEM ANALYSIS AND DESIGN.....	16
4.1 Data Collection Methods.....	16
4.1.1 Interviews	16
4.1.2 Direct Observation	16
4.1.3 Questionnaire.....	16
4.2 System Functionalities and Requirements.....	17
4.2.2 Functional Requirements.....	17
4.2.3 Non-Functional Requirement	19
4.3 Use Case Analysis and Diagram.....	21
4.3.1 Use Case Analysis.....	21

4.3.2	Use Case Diagrams	26
	Figure 3: Customer Use Case Diagram	26
	Figure 4: Owner Use Case Diagram	26
	Figure 5: Property Manager Use Case	27
	Figure 6: Administrator Use Case	27
4.4	Flow chart Diagrams	28
	Figure 7: Customer Flow Chart	28
	Figure 8: Owner Flow Chart	29
	Figure 9: Property Manager	30
	Figure 10: Administrator Flow Chart.	31
4.5	Sequence Diagram	32
	Figure 11: Sequence Diagram of Customer Requesting Property/Unit	32
4.6	Entity Relation Diagram	33
	Figure 12: ER Diagram of PMS	33
4.7	Integrated Diagram	34
	Figure 13: System Users Integrated Diagram	34
4.8	System Architecture Design	35
4.9	Database Design Approach	35
	Figure 14: 3NF Database Schema	36
	REFERENCES	37

LIST OF ABBREVIATION

UDSM	University Of Dar es Salaam
COICT	College Of Information and Communication Technology
RBAC	Role Based Access Control
AWS	Amazon Web Services
JSON	JavaScript Object Notation
REST	REpresentational State Transfer
PMS	Property Management System
3NF	Third Normal Form
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure

CHAPTER ONE

INTRODUCTION

1.1. General Introduction

The real estate and property management sector plays a critical role in economic development and social welfare by providing organized housing, commercial spaces, and related services. In Tanzania, property owners and tenants increasingly face challenges related to trust, transparency, operational efficiency, and access to accurate information. Traditional property management methods rely heavily on manual processes, physical documentation, middle men, and fragmented communication channels, which often result in inefficiencies, delays, data inconsistencies, and increased operational costs.

With the advancement of information and communication technologies, there is a growing need to digitize property management operations to enhance service delivery, improve transparency, and build trust between landlords, property managers (caretakers), and tenants. Digital platforms can integrate leasing, payments, maintenance, communication, and analytics into a single system, offering a more reliable and scalable solution.

This project proposes the design and development of a comprehensive digital property management system that automates tenant onboarding, lease management, payments, maintenance operations, and communication.

1.2. Statement of the Problem

Many housing institutions in Tanzania still rely on manual and paper-based systems for tenant onboarding, property allocation, lease management, and record keeping. A practical example of this challenge can be observed with current property owners and caretakers, where prospective tenants are required to physically visit the property/Unit by contacting the middleman, inspect the house and proceed with manual processes of signing the leasing contract and pay the middleman which increase expenses. This also includes property owners and caretakers having to manually keep track of leases and maintenance requests.

This manual approach presents several problems:

- Delays in tenant onboarding and property allocation
- High risk of data loss, duplication, and human error
- Limited transparency in property availability and tenant selection

- Difficulty in tracking tenant history, payments, and lease status
- Increased operational monitoring and maintenance efforts on homeowners and caretakers
- Weak communication between tenants and property managers

As the number of tenants and properties grows, the existing manual processes become increasingly inefficient and unsustainable. There is a clear need for a centralized, digital, and automated system that can streamline tenant onboarding, manage leases, track payments, and enhance trust and accountability across all stakeholders.

1.3. Objective

1.3.1. Main Objectives

The main objective of this project is to design and develop a digital property management system that automates tenant onboarding, lease management, financial transactions, maintenance operations, and trust verification in order to improve efficiency, transparency, and reliability in property management.

1.3.2. Specific Objectives

The specific objectives of this project are to:

1. Digitize tenant onboarding and lease management processes.
2. Provide a centralized database for tenants, properties, landlords and caretakers.
3. Automate rent collection, expense tracking, and financial reporting.
4. Implement a maintenance request and tracking system for tenants and property managers.
5. Enable automated alerts/reminders through messages.
6. Provide visual property representation including location, status, and dimensions.
7. Implement role-based access control for secure system usage.
8. Introduce trust and verification mechanisms for landlords and tenants.

1.4. Significance of the Project

This project is significant to various stakeholders:

- Property Owners and Property Managers: Enables effective management of multiple properties, transparent tenant selection, and automated financial operations.
- Tenants: Simplifies application processes, improves access to property information, and enhances trust and communication.
- Government and Regulatory Bodies: Supports improved compliance, record keeping, and transparency in the housing sector.

1.5. Project Scope

The scope of this project covers:

- Digital tenant onboarding and lease life-cycle management
- Financial management including rent collection and reporting
- Maintenance request handling and vendor management
- Communication and notification systems
- Property visualization and status representation
- Security features such as RBAC, audit logs, and document storage
- Trust and verification mechanisms for landlords and tenants

The project focuses on software design and implementation, and does not include physical property inspection or external legal enforcement mechanisms.

1.6. Organization of the report

This report is organized into the following chapters:

- Chapter One: Introduction, problem statement, objectives, significance, scope, and report structure.
- Chapter Two: Literature review and related works.
- Chapter Three: Methodology.
- Chapter Four: System analysis and architectural design.

CHAPTER TWO

LITERATURE REVIEW

2.1. Overview of Property Management Systems

Property management systems (PMS) are widely used to automate leasing, payments, tenant communication, and maintenance operations. Commercial systems such as Buildium, AppFolio, and Yardi are commonly cited in literature as mature, feature-rich solutions supporting landlords and property managers in developed economies. These platforms provide integrated modules for tenant screening, online payments, maintenance tracking, and financial reporting.

However, a critical review reveals that these systems are designed primarily for private real estate markets operating within well-established digital identity, banking, and legal infrastructures. For example, Buildium relies on credit bureaus, Social Security Numbers, and card-based payment systems that are not directly applicable in the Tanzanian context.

Other plausible Property management System based on Tanzania is Homes Be-Foward which basically acts as a middleman hosting a real estate marketplace connecting owners and buyers, lacking many key functionalities to help homeowners manage and service their tenants effectively.

Per research, there seem to be a lacking of any substantial Property Management System reliable for Residential purposes as homeowners simply rely on middlemen and their manual efforts.

This gap highlights the need for context-aware property management systems that address both operational efficiency and customer demand.

2.2. Tenant Onboarding and Identity Verification: Manual Processes vs Digital Integration

Tenant onboarding is a critical process that determines eligibility, trust, and long-term tenancy outcomes. In the current scheme where middlemen are hired to seek customers for the home owners, onboarding is conducted by middlemen using social media and other fractured communication channels to post available homes with low successions and often requires possible customers to physically visit the homes. This method does face the scrutiny of deception and forgery where malicious people can impersonate the owner of a certain house to steal money from innocent citizens.

2.3. Financial Management and Payments: Foreign Gateways vs Tanzanian Payment Ecosystem

Financial management is central to property administration, particularly for installment-based housing schemes such as those operated by Property Owners/Landlords. The Tenant Purchase Scheme requires structured down payments and long-term monthly installments, which are currently tracked manually through bank confirmations.

Commercial PMS platforms support online rent payments via credit cards, ACH transfers, or PayPal. While effective in developed economies, these mechanisms do not align with Tanzania's dominant payment infrastructure, which includes:

- Mobile money platforms (M-Pesa, Tigo Pesa, Airtel Money)
- SIM-banking services

The lack of native support for SIM-banking and mobile money in international PMS solutions results in poor adoption and manual workarounds. Literature on financial digitization in developing economies emphasizes that systems must align with local payment behavior and regulatory frameworks to be effective. So, we shall aim to develop a framework that is ready to be integrated with such financial systems in the future.

Consequently, a localized property management system that supports Tanzanian payment gateways is more suitable than importing foreign software designed for different financial ecosystems.

2.4. Maintenance, Communication, and Operational Systems

Maintenance management and tenant communication significantly influence tenant satisfaction and asset longevity. Commercial PMS solutions provide digital ticketing systems, automated vendor dispatch, and tenant portals for issue tracking.

However, in local/Residential property management, maintenance requests are often managed informally through phone calls. While foreign PMS platforms offer advanced features, their deployment assumes:

- High digital literacy among tenants
- Continuous internet access
- Private contractor ecosystems

These assumptions do not always hold in Tanzanian public housing environments. Furthermore, existing systems do not integrate maintenance workflows with credible and reliable maintenance principles and regulatory procedures.

This mismatch between feature design and operational reality explains why many advanced systems fail to deliver expected value when adopted without localization. A system designed specifically for Tanzanian public use must balance digital efficiency with practical usability and regulatory compliance.

2.5. Related Works

Several property management systems have been developed globally to address challenges related to leasing, payments, and maintenance. Commercial International platforms such as Buildium, AppFolio and Rentec Direct offer integrated solutions for private landlords and property managers. However, these systems are often designed for developed markets and may not align with Tanzanian regulatory, financial, and identity verification requirements.

Academic studies on housing management systems in developing countries highlight challenges such as limited digitization, lack of trust mechanisms, and poor integration with national identity systems. Few systems incorporate tenant reputation scoring, landlord verification, or predictive analytics tailored to local contexts.

Compared to these existing national works, Homes Be Forward represents an incomplete and middleman-based system solution and the local Property Owner, Property Manager and middleman working scheme representing a rigid and manual system. While functional, they lack automation, real-time access, and advanced decision-support features. This comparison highlights the need for a localized, context-aware solution that addresses both operational and trust-related challenges.

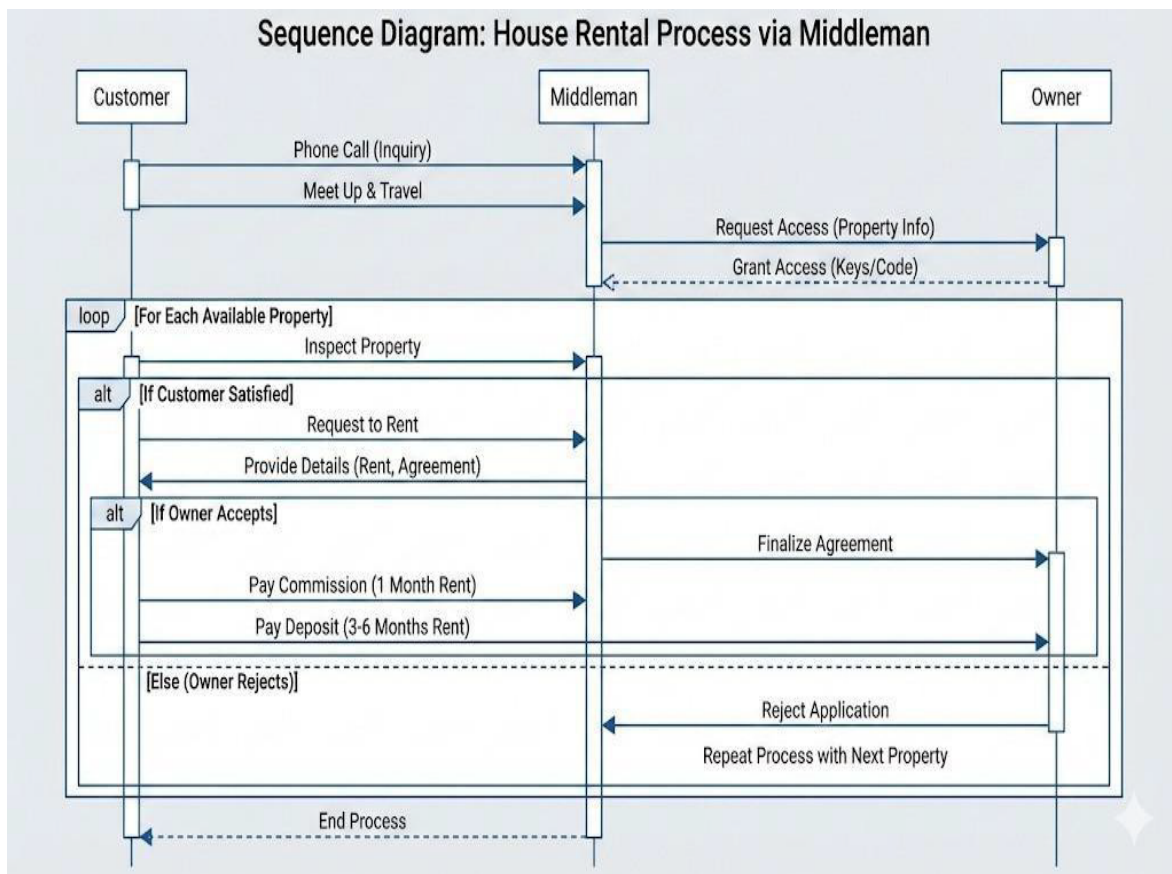


Figure 1: House Rental Process via Middleman at Current residential processes.

2.6. Weaknesses of Existing Property Management Systems

The current traditional Property Management system, the local Property Owner, Property Manager and middleman working scheme, to some degree has been working for the current populous does have the following weaknesses:

- Automation of rent collection, expense tracking, and financial reporting.
- Digitized tenant onboarding and lease management processes.
- A centralized database keeping track and history of tenants, properties, landlords and property managers.

2.7. Project Gap

Current majority property management practices in Tanzania rely heavily on manual workflows, physical documentation, and physical-based interactions. There is no end-to-end digital tenant onboarding process, limited automation in financial management, and insufficient mechanisms for maintenance tracking and communication.

Additionally, existing systems lack trust and verification features such as landlord verification badges, tenant reputation scoring, and centralized document vaults. Advanced analytics, including predictive maintenance and tenant behavior analysis, are largely absent.

This project seeks to bridge these gaps by proposing a comprehensive digital property management system tailored to the Local population i.e. Landlords, Caretakers and Tenants. By integrating automation, verification, communication, and analytics, the system aims to modernize property management practices and improve service delivery for all stakeholders.

CHAPTER THREE

METHODOLOGY

3.1. Research Design and Approach

This project adopts an applied research approach focused on the design and development of a practical software solution to address real-world challenges in property management. The research is solution-oriented, aiming to digitize and automate tenant onboarding, lease management, financial operations, maintenance processes, and trust verification within the Tanzanian housing context.

The study follows a system development–based research design, where existing processes are analyzed, system requirements are derived, and a functional software system is designed and implemented.

3.2. System Development Methodology

3.2.1 Development Methodology

The system is developed using the Agile software development methodology. Agile supports incremental development through short iterations, allowing continuous refinement of features and early validation against requirements. This approach is suitable for the project because system requirements may evolve as implementation progresses and stakeholder feedback is incorporated.

Each iteration focuses on implementing specific system modules such as onboarding, payments, maintenance, or document management.



Figure 2: Agile Development workflow

3.2.2 Scrum Agile Framework

Within the Agile methodology, the Scrum framework was adopted to structure the development process into manageable, iterative cycles known as sprints. Each sprint spans one week, allowing for frequent review and incremental delivery of working features.

The Scrum process includes the following key components:

- A. **Sprint Planning:** At the beginning of each sprint, tasks are identified and prioritized in collaboration with the team, focusing on delivering a functional subset of the system features.
- B. **Daily Standups:** Short daily meetings are held to discuss progress, address challenges, and realign the team's efforts as needed.
- C. **Sprint Review:** At the end of each sprint, the completed work is demonstrated to supervisor and stakeholders for feedback.
- D. **Sprint Retrospective:** The team reflects on the sprint to identify areas of improvement for the next cycle.

This structure provides a flexible approach and ensures that the development process remains efficient and user-centered. Each sprint contributes to delivering a functional requirement application capable of addressing the challenges faced by the UDSM students.

CHAPTER 4

PROPERTY MANAGEMENT SYSTEM: SYSTEM ANALYSIS AND DESIGN

4.1 Data Collection Methods

This study employed a mixed-method data collection approach to ensure comprehensive understanding of property management challenges from both tenants and landlords. The methods combined qualitative techniques such as interviews and direct observation to explore operational and trust-related issues, and quantitative techniques such as questionnaires to validate recurring problems and user needs. This approach enabled the identification of real workflow inefficiencies, communication gaps, payment tracking challenges and system usability expectations, forming a strong foundation for system requirements analysis and design.

4.1.1 Interviews

Semi-structured interviews were conducted with selected tenants and landlords to gain in-depth insights into existing property management practices. The interviews revealed key issues including manual rent tracking, poor record keeping, lack of transparency in tenant verification, delayed maintenance handling and communication breakdowns between stakeholders. The impact of this method was the identification of critical functional requirements such as digital onboarding, automated rent tracking, maintenance ticketing and trust verification mechanisms.

4.1.2 Direct Observation

Direct observation was used to examine how property management activities are currently executed in real environments. It was observed that payments are often recorded manually in notebooks, maintenance requests are handled informally via phone calls or messaging applications and lease documents are stored physically without centralized tracking. This method highlighted process inefficiencies, risk of data loss and lack of accountability, which directly influenced the need for centralized database management, audit logs and structured workflow automation in the proposed system.

4.1.3 Questionnaire

Structured questionnaires were administered to tenants, particularly students renting houses, to collect measurable data on common challenges and system expectations. The responses indicated frequent issues such as unclear rent records, delayed maintenance response, limited visibility of property availability and

preference for mobile money payment options. The impact of this method was the statistical validation of system features, especially online payment integration, maintenance request tracking, automated notifications and user-friendly web access.

4.2 System Functionalities and Requirements

This section presents the core functionalities of the proposed Property Management System (PMS) and formally defines the functional requirements derived from the data collection findings and system objectives.

4.2.1 Core System Functionalities

The proposed Property Management System shall provide the following core functionalities:

- User registration and authentication
- Role-Based Access Control (RBAC)
- Property listing, visualization and management
- Digital tenant onboarding and application processing
- Lease creation and management
- Rent payment recording and tracking
- Maintenance request submission and tracking
- Notification and alert system
- Secure document storage

4.2.2 Functional Requirements

According to Dennis, Alan, Barbara Haley Wixom, and Roberta M. Roth (2012), Systems Analysis and Design, 5th ed., Wiley, functional requirements are the requirements that describe what the software or the system should do. Functional requirements are categorized into two types.

i Evident Functions whereby the requirement will be performed and recognized by the user.

ii Hidden Functions whereby the requirement will be performed by the system, but not visible to users.

Ref No	Functional Description		Category
F1.	USER MANAGEMENT		
	F1.1	The system shall allow users, tenants, landlords, caretakers, administrators to register and log in securely.	Evident
	F1.2	The system shall enforce Role-Based Access Control to restrict access based on user roles.	Evident

	F1.3	The system shall allow administrators to manage user accounts.	Hidden
F2.	PROPERTY MANAGEMENT		
	F2.1	The system should allow landlords caretakers to create and manage property listing	Evident
	F2.2	The system should allow updating property details including rent amount, location and availability status.	Evident
F3.	TENANT ONBOARDING AND LEASE MANAGEMENT		
	F3.1	They should allow tenants to view available properties	Evident
	F3.2	The system should allow tenants to submit digital rental applications.	Evident
	F3.3	The system should allow landlords or caretakers to review and approve or reject tenant applications.	Evident
	F3.4	The system should allow creation of lease agreements specifying rent amount, duration, and terms.	Evident
	F3.5	The system should track lease start date and expiry date.	Evident
	F3.6	The system should generate lease expiry notifications.	Hidden
F4.	FINANCIAL MANAGEMENT		
	F4.1	The system should record rent payments made by tenants.	Evident
	F4.2	The system should track outstanding balances and overdue payments.	Evident
	F4.3	The system should generate rent due reminders before payment deadlines.	Evident
	F4.4	The system should maintain payment history for each tenant.	Evident
	F4.5	The system should generate financial summary reports for landlords.	Hidden
F5.	MAINTENANCE MANAGEMENT		
	F5.1	The system should allow tenants to submit maintenance requests digitally.	Evident
	F5.2	The system should allow landlords or caretakers to update maintenance request status.	Evident
	F5.3	The system should allow tracking of maintenance status such as Submitted, In Progress or Resolved.	Evident
	F5.4	The system should maintain maintenance history records for each property.	Hidden
F6.	DOCUMENT MANAGEMENT		

	F6.1	The system should allow uploading of lease agreements and verification documents.	Evident
	F6.2	The system should restrict document access based on user roles.	Hidden
	F6.3	The system should securely store documents in a centralized repository	Hidden

4.2.3 Non-Functional Requirement

According to Dennis, Alan, Barbara Haley Wixom, and Roberta M. Roth (2012), Systems Analysis and Design, 5th ed., Wiley, Non-functional requirements are the characteristics that the system should have. These are the qualities, standards, or criteria that the system must meet to function effectively. These requirements mostly focus on reliability, performance, scalability, usability, and maintainability. These are called attributes of Non-functional requirements.

Table 4.2 shows the Non-functional requirements for the Property Management System

Attribute	Non-Functional Requirement
SECURITY	The system should ensure that all user passwords are securely encrypted before being stored in the database.
	The system should enforce secure authentication mechanisms to prevent unauthorized access.
	The system should restrict system access based on user roles and permissions.
	The system should maintain audit logs of critical system activities
PERFORMANCE AND EFFICIENCY	The system should respond to user requests within an acceptable time
	The system should handle multiple users accessing the system concurrently without performance degradation.
	The system should efficiently process rent tracking and maintenance updates without data inconsistency.
RELIABILITY AND AVAILABILITY	The system should ensure data integrity during transactions such as rent payments and lease creation.
	The system should maintain regular database backups to prevent data loss.
	The system should be available to users at all times except during scheduled maintenance.
USABILITY	The system should provide a user-friendly web

	interface accessible through standard web browsers.
	The system should provide clear navigation menus and structured dashboards for each user role.
	The system should provide clear error messages and confirmation notifications.
SCALABILITY AND MAINTAINABILITY	The system should be scalable to support increasing numbers of users and properties.
	The system should be designed using modular architecture to allow future feature expansion.
	The system should support integration with external payment gateways such as mobile money services.
PORTABILITY	The system should be accessible via modern web browsers without requiring additional software installation.
	The system should be deployable in a cloud environment such as AWS.

4.3 Use Case Analysis and Diagram

4.3.1 Use Case Analysis

01. SIGNING IN / REGISTRATION

USE CASE NAME	SIGN UP OR REGISTRATION
ACTORS	Customer, Owner, Property Manager, Administrator
DESCRIPTION	The system should allow Customers/Possible Tenants, Owners, Property Managers and Administrators to create accounts.
PRECONDITIONS	1. The user should have relevant information about the required filled-in information. 2. The user's device should be connected to the internet
MAIN FLOW	Create an Account: 1. User enters details (name, email, username, password etc.) 2. The system makes users crosscheck the data they are delivering. 3. The system verifies the data. 4. Account is created 5. System confirms access
POSTCONDITION	The user is registered and redirected to the respective page.
EXCEPTIONS	1. If email or username is already registered, show an error 2. If the data is invalid i.e. not the right data type, show the error

02. LOGIN

USE CASE NAME	LOGIN
ACTORS	Customer, Owner, Property Manager, Administrator
DESCRIPTION	Authenticates users to access the system.
PRECONDITIONS	1. The user's device should be connected to the internet 2. The user should have an account in the system.
MAIN FLOW	1. User enters username or email & password. 2. System verifies credentials 3. The user is logged in.
POSTCONDITION	1. The user is logged into the system 2. The user should be directed to the respective page
EXCEPTIONS	If the credentials do not match the database of registered accounts: 1. Throw an error requesting the user to enter the right credentials 2. If they aren't registered, offer the option to register an account.

03. UPLOAD PROPERTY

USE CASE NAME	UPLOADING A PROPERTY
ACTORS	Owner
DESCRIPTION	The system should allow a registered Property Owner to upload their described property to the system for customers to view and request occupation
PRECONDITIONS	1. The Owner must be registered and logged in 2. The user's device should be connected to the internet 3. The owner must have all necessary information about the property i.e. property location, rent, property dimensions and other property details
MAIN FLOW	1. Owner enters property details (location, rent etc.) 2. The system requests the owner to crosscheck and verify the submitted information. 3. The system verifies and saves the data. 4. Property is registered.
POSTCONDITION	The property is registered and is visible and accessible to potential tenants/customers.
EXCEPTIONS	1. If the property is already registered, throw an error and show the option to view the registered property. 2. If the data is invalid i.e. not the right data type, show the error

04. PLACING OFFERS FOR AVAILABLE PROPERTY/UNIT.

USE CASE NAME	PLACING OFFERS FOR AVAILABLE PROPERTY/UNIT.
ACTORS	Customer
DESCRIPTION	The system shows customers available properties. This also includes booked and occupied units in the same property.
PRECONDITIONS	1. The user's device should be connected to the internet.
MAIN FLOW	1. Customer enters the system domain name to access the web page 2. The system provides a short description of the business process to guide customers on how to navigate the webpage. 3. Customer browses through properties of his/her desired generic location 4. The customer selects/puts up an offer for a property. 5. The customer has to log in to the system to make the request. If they have no registered account to the system, they are to register an account.
POSTCONDITION	A customer sends a request for the property to the Property Manager
EXCEPTIONS	1. If customer selects a booked/occupied unit/property, deny request 2. If user makes more than 1 request for the same property/unit, throw an error

05. REVIEWING PROPERTY REQUEST/OFFERS FROM CUSTOMERS

USE CASE NAME	REVIEWING PROPERTY/UNIT REQUESTS
ACTORS	Property Manager, Owner.
DESCRIPTION	The system first informs the property manager of requests for the unit/property from various customers. He/She shall send requests found favorable to the Owner to decide who gets to book and/or occupy the property
PRECONDITIONS	1. The Property Manager and Owner are registered and logged in the system. 2. The user's device should be connected to the internet
MAIN FLOW	1. Property Manager reviews offered requests for unit/property occupation. 2. Property Manager selects favorable requests offered to the Owner. 3. The owner reviews requests sent by the Property Manager. 4. The owner accepts one Customer Request offer and sends feedback to the Property Manager. 5. The property informs the selected customer that their requests are accepted. The rest are notified that their requests are denied.
POSTCONDITION	1. The property status changes from available to booked. 2. The customer receives the status of his request offer.
EXCEPTIONS	1. If the owner attempts to accept more than 1 request for the same unit/property, throw an error.

06. LEASE PAYMENT

USE CASE NAME	LEASE PAYMENT
ACTORS	Customer, Owner
DESCRIPTION	The system gives the customer payment details he/she needs to make the relevant lease payment to become an official tenant of the selected unit.
PRECONDITIONS	1. The Owner has provided the updated and valid payment details 2. The Customer has the full amount of money to pay the rent in full. 3. Both the Owner and the Customer have signed the contract with all agreements well documented, reviewed and accepted by both parties.
MAIN FLOW	1. The customer takes payment details of the owner. 2. The customer makes payment. 3. The system picks up on the payment and notifies the owner. 4. The owner crosschecks his/her banking or mobile money accounts and verifies payment.
POSTCONDITION	1. The property is updated from booked to occupied property/unit 2. The customer is now granted access to the property/unit

	3. The customer is now a tenant to the property manager and owner.
EXCEPTIONS	1. If payment is not complete, the owner is notified of the incomplete payment and has to communicate with the customer on what is to proceed: a. The owner might grant access to the property if further agreements are made where future payments can be made b. The owner may choose to reject incomplete payments, return the money back to the customer and terminate contract. This would demote the property status back to available.

07. PLACING AND SERVICING MAINTENANCE REQUEST

USE CASE NAME	PLACING AND SERVICING MAINTENANCE REQUEST
ACTORS	Customer/Tenant, Property Manager and Owner
DESCRIPTION	The system offers a functionality for the tenant to send maintenance requests when some problems occur to the Property Manager. The Property Manager then relays information to the owner for next stages to proceed.
PRECONDITIONS	2. The customer is now a tenant to the system.
MAIN FLOW	6. Property Manager first receives the maintenance requests from the tenant. 7. Property Manager then relays details of the maintenance request to the Owner for it to be dealt with accordingly. 8. The owner sends feedback to the property manager. This involves but not limited to: a. Funds request from the tenant to cover some costs. b. Funds to cover the repair costs. c. Accepted technician, requested by the property manager, to undertake the task 9. The maintenance request is then serviced according to time and money constrains 10. The tenant then sends feedback to the property manager on the entire maintenance request servicing procedures.
POSTCONDITION	The maintenance request is worked on to completion.
EXCEPTIONS	2. If the tenant attempts to accept more than 1 request for the same maintenance request, throw an error.

08. REPORT SYSTEM ISSUES

USE CASE NAME	REPORT SYSTEM ISSUES
ACTORS	Customer/Tenant, Property Manager, Owner and Administrator
DESCRIPTION	The system offers a functionality for system users i.e. owners, customers/tenants and property managers to report system issues that should be resolved by the administrator and the development team.
PRECONDITIONS	1. The customers/tenants, property owners and property managers have access to the system according to their privileges. 2. The non-administrative user has detected a bug/issue that the system inhibits them from certain uses of the system.
MAIN FLOW	1. A non-administrator sends their system issue report to the right portal handling issues. 2. The administrator receives them and attempts to resolve some by him/her self. 3. Those which need more expertise they are sent to: i. Development team ii. System testers 4. The issue is then worked on to completion or resolution (may be put on hold depending on its severity)
POSTCONDITION	The reported system issue is worked on to completion or resolution
EXCEPTIONS	The system should deny any administrative user to file the same exact issue more than once

09. MANAGE USERS

10.

USE CASE NAME	MANAGE USERS
ACTORS	Administrator
DESCRIPTION	The system offers a functionality for administrators to manage users registered to the system.
PRECONDITIONS	1. The administrator is registered and logged in to the system. 2. There are users that wish to edit or fully terminate their accounts from the system. 3. There may be some malicious activities being done by some users.
MAIN FLOW	1. The administrator located the relevant registered accounts. 2. He/she makes the necessary changes: a. Edit some crucial details b. Delete account
POSTCONDITION	The state of registered account changes.
EXCEPTIONS	If user not found, throw an error.

4.3.2 Use Case Diagrams

- Customer:

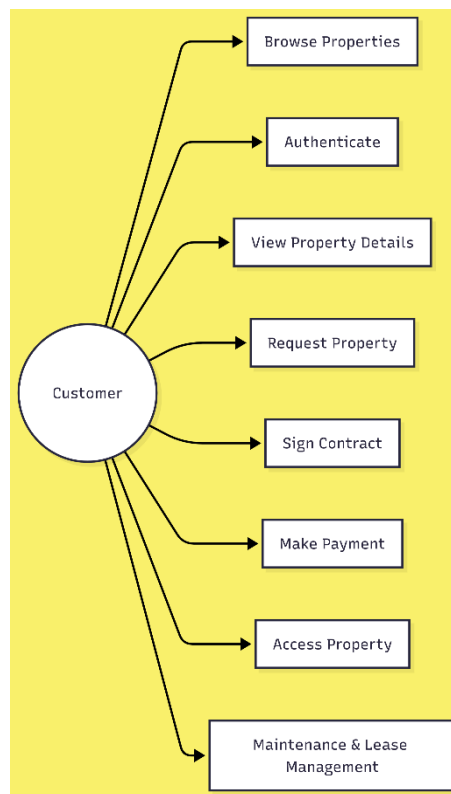


Figure 3: Customer Use Case Diagram

- Owner:

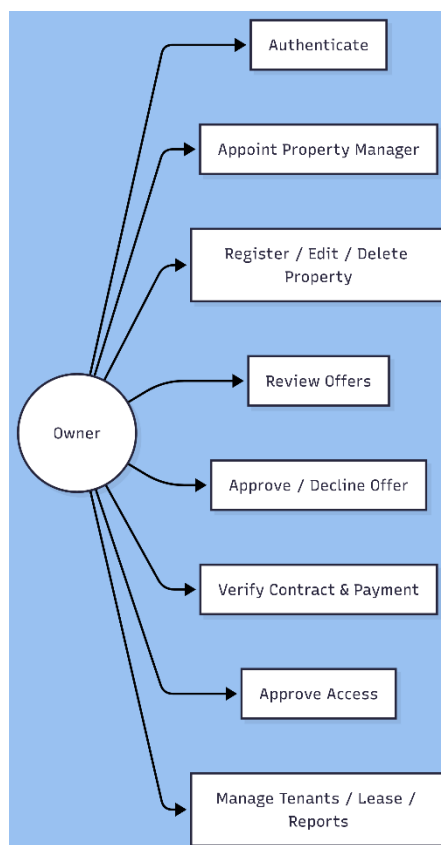


Figure 4: Owner Use Case Diagram

- Property Manager:

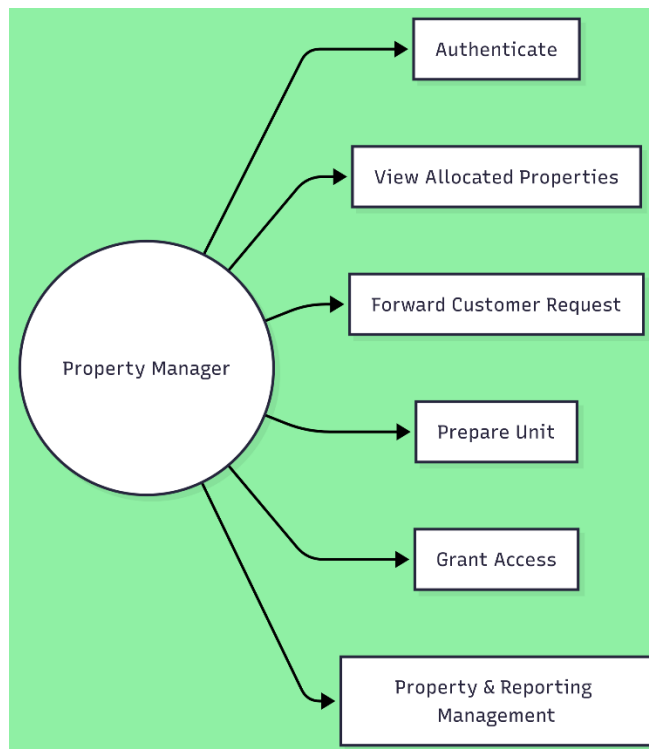


Figure 5: Property Manager Use Case

- Administrator:

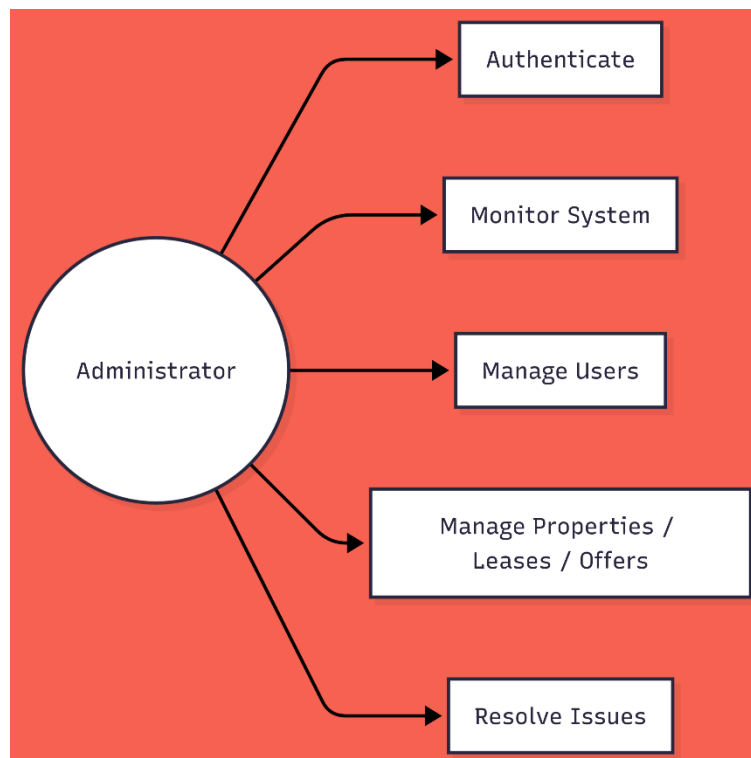


Figure 6: Administrator Use Case

4.4 Flow chart Diagrams

4.4.1 Customer Flow Chart:

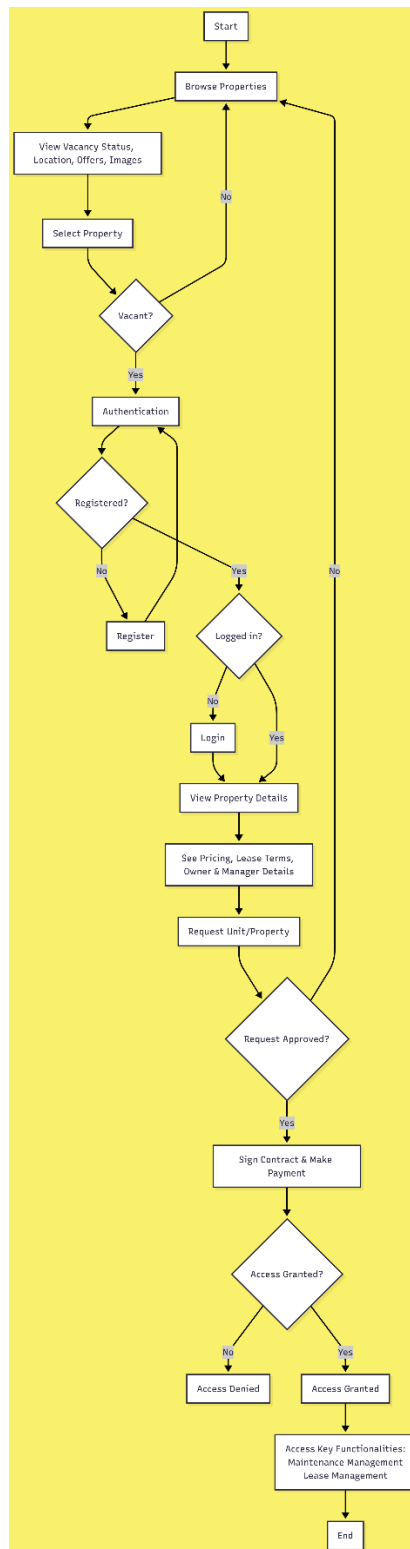


Figure 7: Customer Flow Chart

4.4.2 Owner Flow Chart

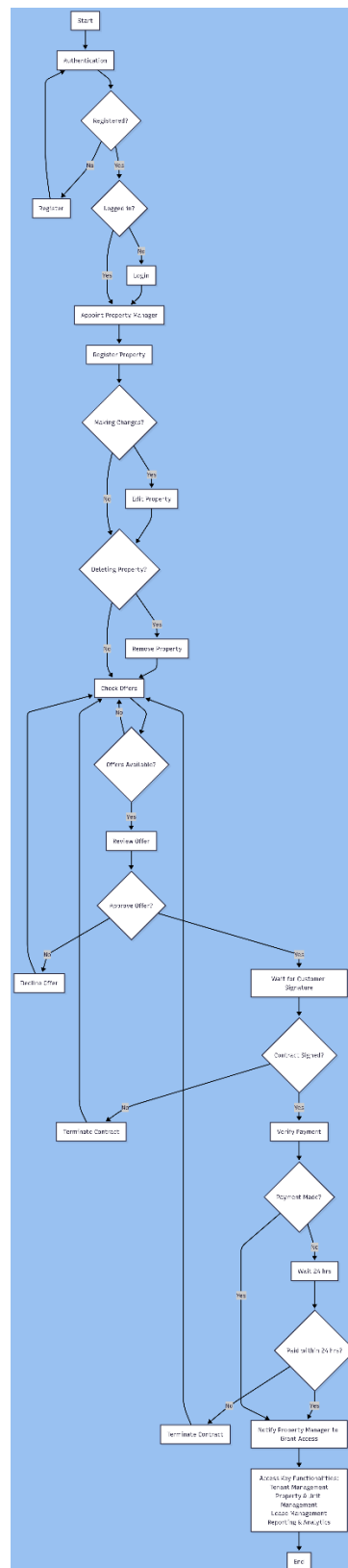


Figure 8: Owner Flow Chart

4.4.3 Property Manager:

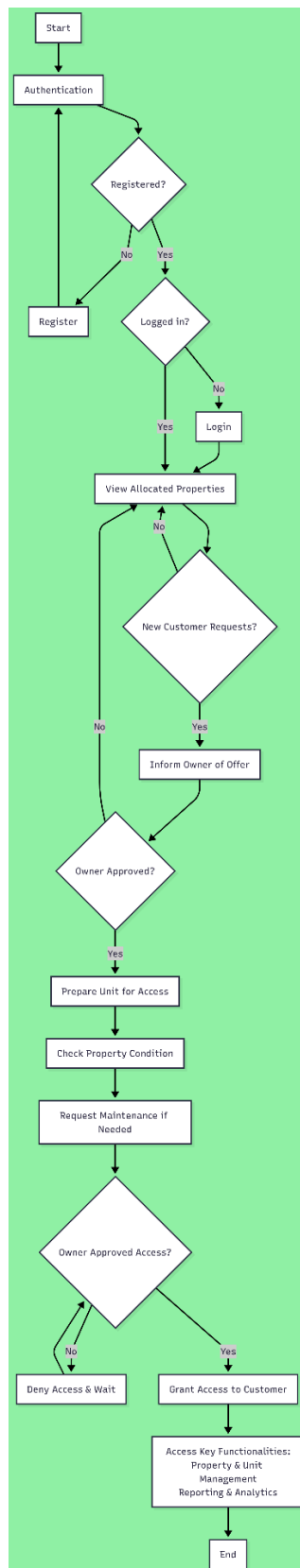


Figure 9: Property Manager

4.4.4 Administrator

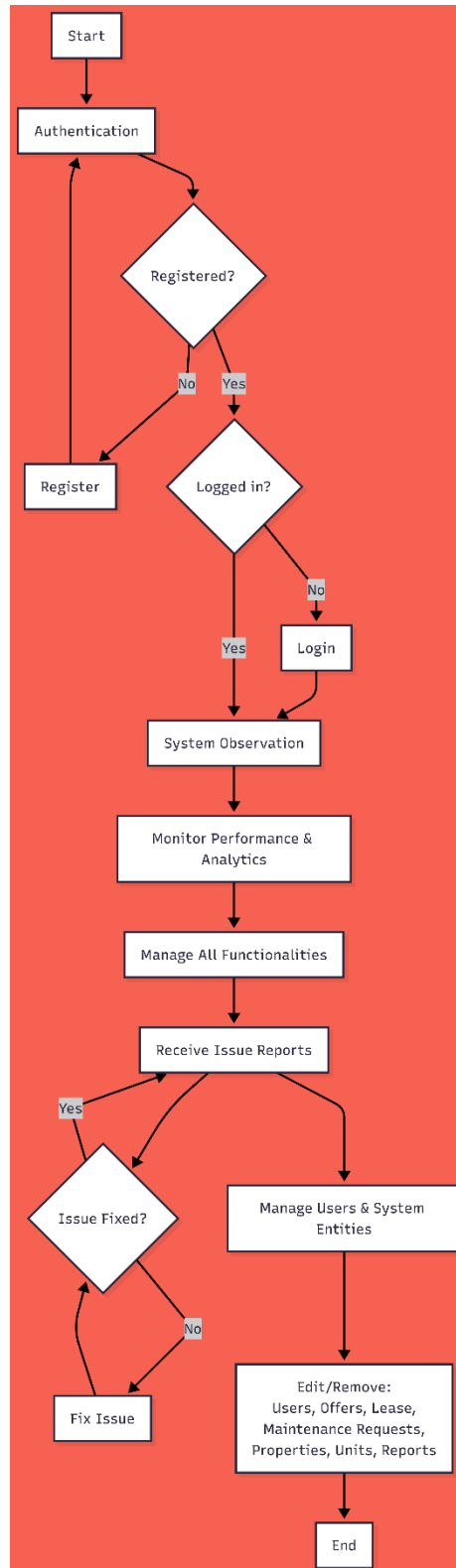


Figure 10: Administrator Flow Chart.

4.5 Sequence Diagram

Customer Process:

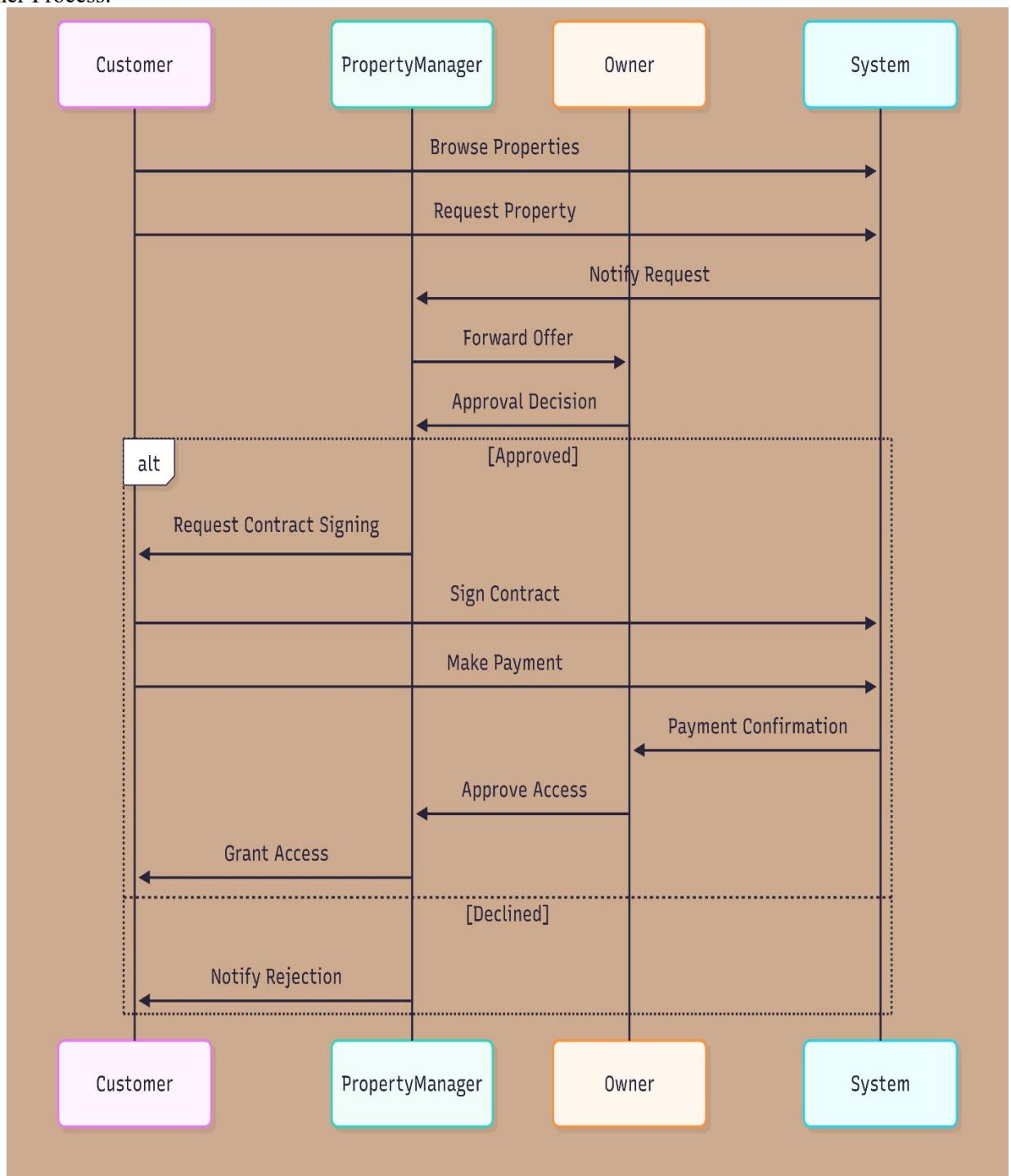


Figure 11: Sequence Diagram of Customer Requesting Property/Unit

4.6 Entity Relation Diagram

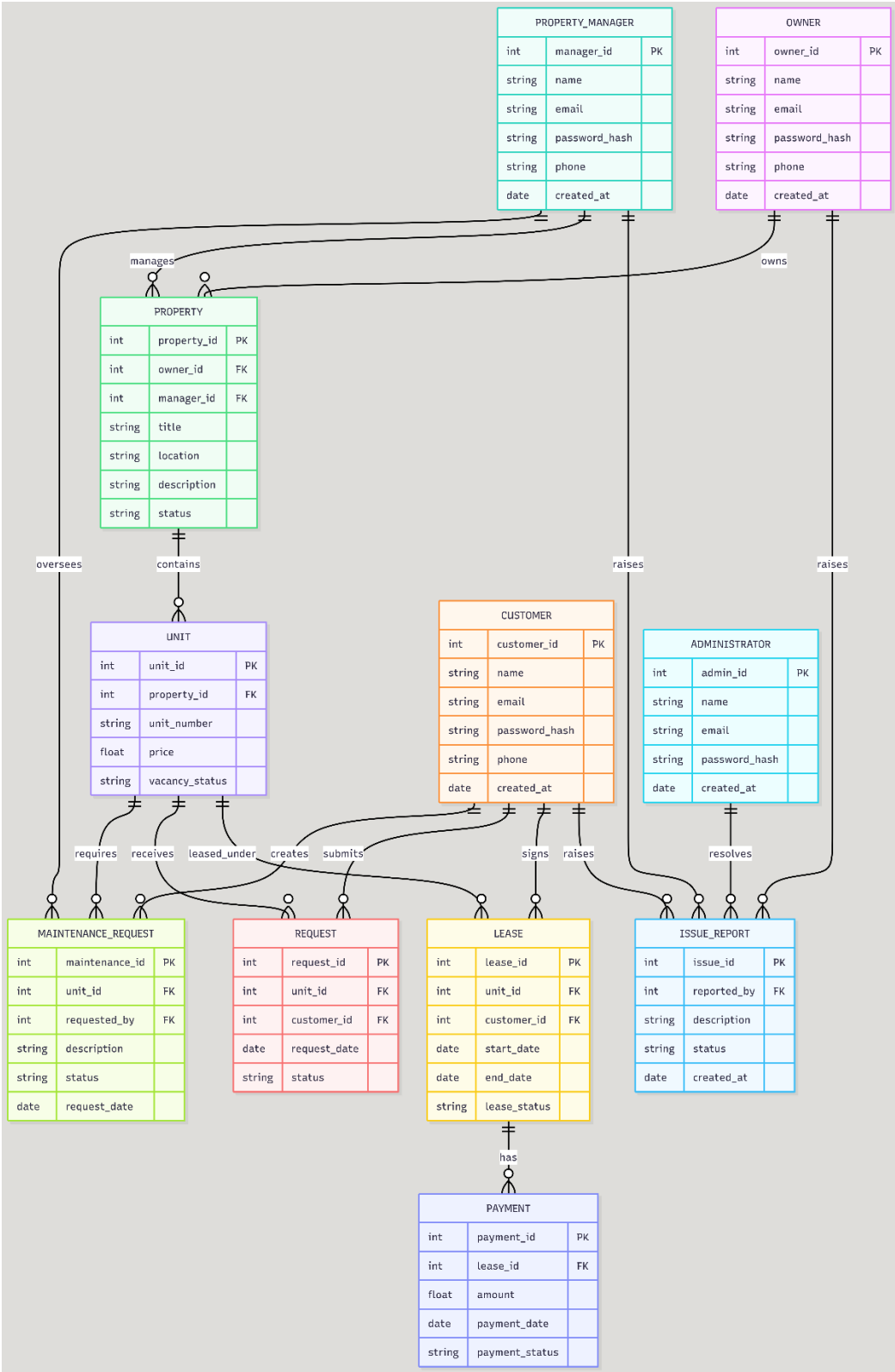


Figure 12: ER Diagram of PMS

4.7 Integrated Diagram

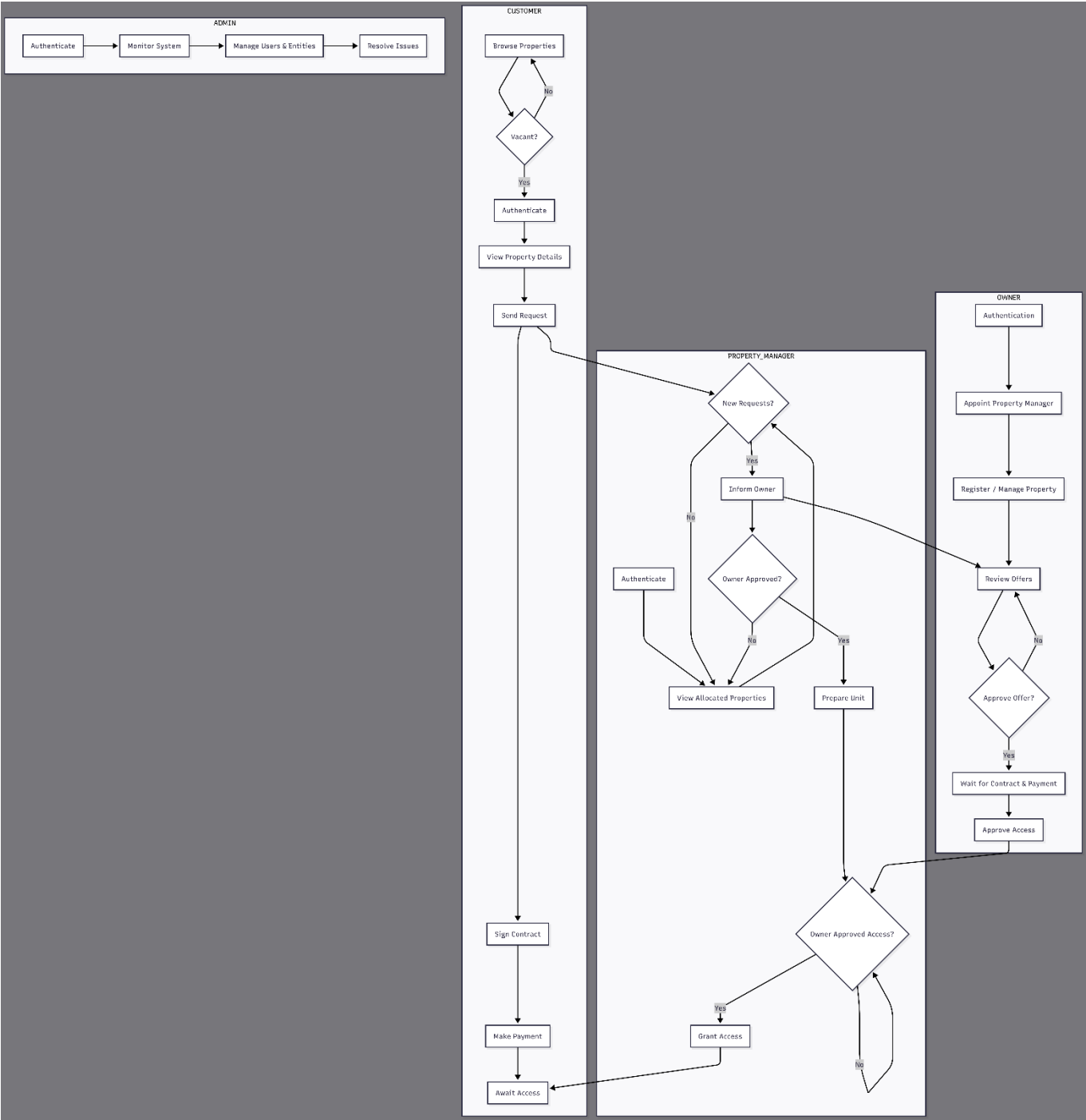


Figure 13: System Users Integrated Diagram.

4.8 System Architecture Design

The system is designed using a client–server architecture implemented through RESTful APIs. This architectural style separates the frontend (client) from the backend (server), enabling independent development, scalability, and easier integration with external systems.

- The client layer is a web-based frontend developed using React, accessed through standard web browsers.
- The application layer consists of a backend developed using Java and the Spring Boot framework, exposing REST APIs for all system operations.
- The data layer is a centralized relational database that stores structured system data.
- The system is designed for cloud deployment using Amazon Web Services (AWS).

Communication between the client and server is conducted over HTTP/HTTPS using JSON-formatted requests and responses.

4.9 Database Design Approach

The system uses PostgreSQL as the database management system due to its robustness, transactional support, and suitability for complex relational data. The database design follows a relational and normalized approach to ensure data integrity and reduce redundancy.

Key principles applied include:

- Entity-based modeling for core entities such as Users, Properties, Tenants, Leases, Payments, Maintenance Requests, and Documents
- Normalization up to Third Normal Form (3NF)
- Use of primary and foreign keys to enforce referential integrity
- Role-based user modeling to support access control
- Audit logging to track critical system actions
- Extensible schema design to allow future integration with external services such as payment gateways

This design ensures consistent, secure, and efficient data management.

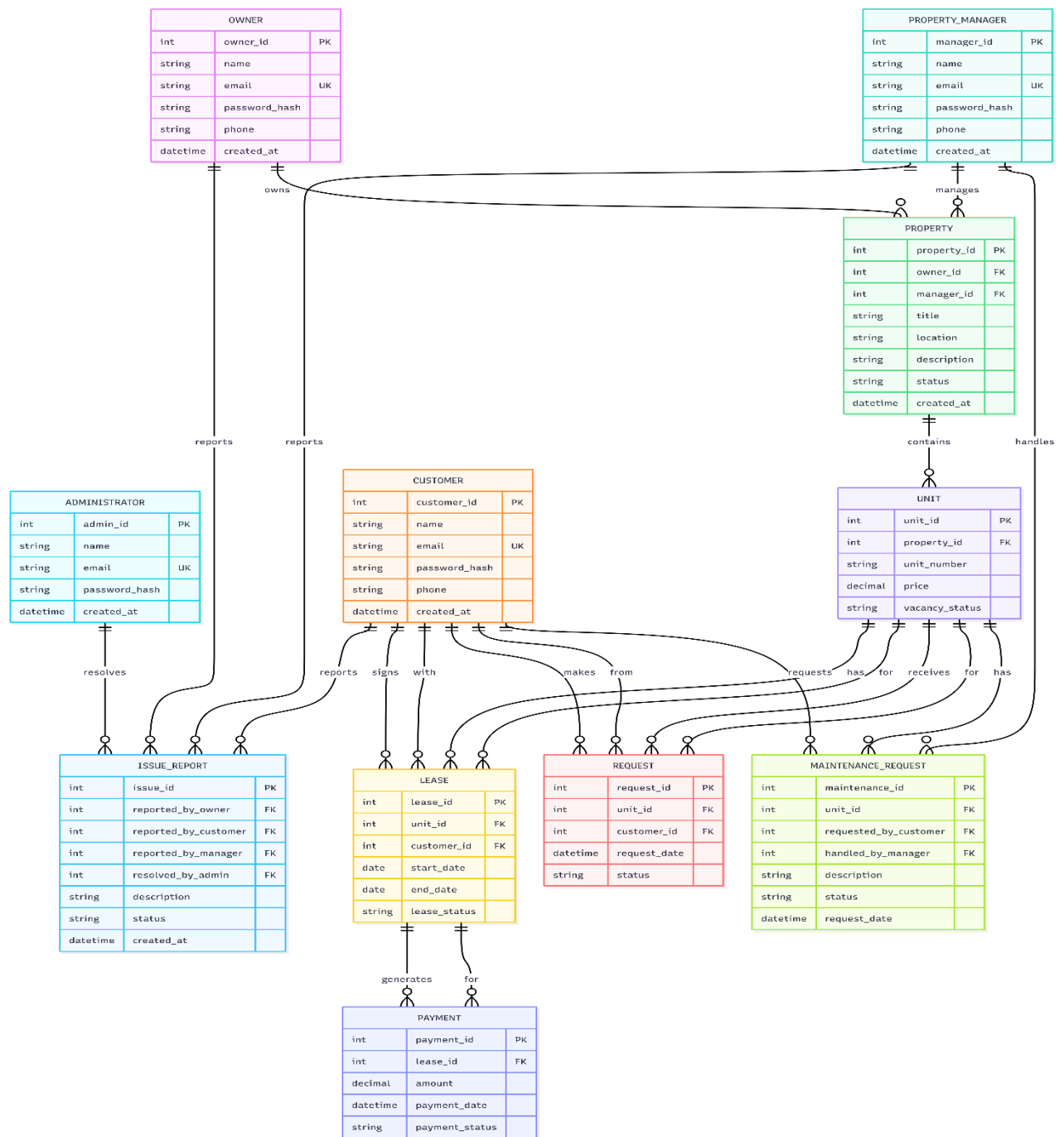


Figure 14: 3NF Database Schema

REFERENCES

<https://fullpdfword.org/reviews/u30FC9/244285/4968440-apartment-management-system-analysis-design>

<https://scand.com/company/blog/how-to-build-property-management-software/>

<https://www.studysmarter.co.uk/explanations/architecture/real-estate/property-management-systems/>

<https://www.taazaa.com/blog/custom-property-management-software-guide>

<https://su-plus.strathmore.edu/server/api/core/bitstreams/fd113d40-d956-434b-9f06-1dcfd55ee6ac/content>

<https://opus.govst.edu/cgi/viewcontent.cgi?article=1534&context=capstones>

<https://www.buildium.com/>

<https://homes.beforward.jp/search/rent/house/all/tanzania>

<https://www.rentecdirect.com/pricing>

<https://www.appfolio.com/>